

Exploring Multitemporal Remote Sensing for Subterranean Feature Detection

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ABSTRACT

This study introduces a remote sensing methodology using multitemporal satellite imagery (Sentinel-2, 2015-2025) to detect a 52-kilometer subterranean water pipeline by analyzing soil marks as surface proxies. The approach involves extracting pixel-level RGB and Near-Infrared (NIR) data from selected areas along and across the pipeline, applying normalization and enhancement techniques to accentuate temporal variations. Results indicate that enhanced analysis of transect datasets successfully highlighted localized indicators of the pipeline's presence. Temporal analysis revealed distinct patterns, including annual periodicity in statistical metrics (RMS, standard deviation) linked to vegetation cycles, and differential pipeline signal discernibility (e.g., clearer post-2020 in Dataset 2; consistent and non-seasonal in Datasets 3 and 4 after enhancement). High-resolution drone imagery (DJI Mavic 3, 3 April 2024) provided preliminary validation, confirming linear soil discolorations (20–25 m wide) aligned with satellite-derived patterns and temporal dynamics. The findings demonstrate the potential of systematic temporal analysis for monitoring underground infrastructure, although detectability is influenced by local pipeline-soil interactions and environmental factors, underscoring the need for in-situ validation. This work aims to form a basis for future systematic pipeline monitoring for leakages and structural integrity.

Keywords: Soil Marks, Satellite Imagery, Temporal Analysis, Leakage Detection.

1. INTRODUCTION

Soil surface marks and particularly referred to discolorations, are perceptible alterations in the inherent color or brightness of the uppermost soil layer compared to adjacent areas, indicative of variations in soil composition, moisture content, or physical properties [1]. These marks can originate from both anthropogenic and natural subsurface phenomena. Natural geological processes also generate soil marks. Oxidation of subsurface ore minerals, particularly sulfides, forms "gossans," which are iron-rich caps appearing as red, orange, or yellow stains due to minerals like goethite and hematite. Anthropogenic activities are significant contributors. The physical mixing of soil horizons during excavation and backfilling introduces contrasting materials; for instance, darker, organic-rich topsoil used as backfill can create dark linear marks, while subsoil like chalk brought to the surface results in lighter marks [2]. Changes in soil properties such as compaction and texture in backfilled areas alter moisture retention, influencing color; less compact, fine grained backfill typically appears darker when moist [2]. Chemical leaching from buried structures, like calcium hydroxide from concrete, can precipitate white calcium carbonate, lightening the soil [2, 3]. Water leakages from pipelines create persistently damp, darker soil patches due to increased moisture content [3].

The study of soil marks is of paramount importance as these surface discolorations function as critical non-invasive indicators of diverse subsurface structures and processes [4]. Their significance is underscored by their utility in archaeological prospection, where subtle variations in soil composition or moisture, resulting from buried features, can reveal the presence of otherwise hidden ancient remains [1]. Furthermore, the monitoring of soil marks is vital for ensuring the integrity and safety of subterranean infrastructure, as changes in soil color or condition can indicate issues such as pipeline leaks, thereby aiding in environmental protection and risk management [5].

Studies examining soil marks, specifically soil discoloration in bare earth, have employed several methods. A fundamental approach is ground-based visual assessment. Archaeologists have historically relied on observing soil color contrasts, such as lighter patches over foundations or dark circles indicating pits [6]. Pipeline inspectors also visually monitor for discolored soil [5]. These visual assessments can be standardized. Aerial reconnaissance and photography

have long been utilized to detect archaeological sites via soil marks[1]. The visibility of soil discoloration caused by subsurface features is significantly influenced by environmental conditions. Key factors modulate whether soil marks are conspicuous:

- **Moisture and Precipitation:** Soil color contrasts are often most apparent after rain followed by drying, creating damp marks that stay darker. Heavy rain can homogenize moisture, reducing contrast. The detectability of damp marks is linked to climate and soil type. Soil moisture content directly impacts reflectance values, with increased moisture causing darker soil appearance.
- **Illumination and Sun Angle:** Low-angle sunlight (e.g., early morning or late afternoon) can enhance subtle textural and tonal differences in soil that are washed out by high noon sun. Optimal sun angles (e.g., 30-45°) have been suggested for aerial photography.
- **Season and Vegetation Cover:** Bare-earth conditions, such as after harvest or before crops grow, are typically required. Droughts can reveal features as areas with better water retention stay moister, creating contrast (parch marks). Moderate stress conditions often yield the best contrasts. The right season is crucial for remote sensing of soil discoloration (Agapiou et al., 2016; Casana, 2020).
- **Soil Background and Composition:** The inherent color and texture of the native soil determine the contrast of an anomaly. High-contrast situations, like dark fill in light soil, are easiest to detect. Grain size affects moisture retention and color when wet.

Studies examining these factors include research on damp marks related to climate and soil type (the influence of precipitation on spectral responses and the impact of weather and time of day/year on water leak detection using thermal/vegetation indice. the temporal nature of these features requires optimally timed observations [1,3,4,6].

To the best of the authors' knowledge, traditional methods for detecting soil marks face limitations in systematically studying their temporal variations. Ground color differences indicating subsurface structures are not always visible throughout the year and may only be observable under specific climatic conditions [4]. This periodicity and variability make them difficult to detect using methods that rely on data from a single point in time or isolated intervals, potentially causing significant traces to be overlooked [4,6]. Systematic study of these temporal variations is crucial because the visibility of soil marks often depends on specific, short-lived "windows of opportunity" determined by environmental dynamics. Understanding these temporal dynamics is essential for planning effective detection strategies, including optimizing the timing of observations to capture transient phenomena [3]. Furthermore, long-term monitoring and analysis of these changes can help in distinguishing between natural and anthropogenic causes, identifying active processes, and revealing patterns that sporadic observations would miss.

This study aims to conduct a systematic, long-term investigation of soil surface traces observed above a subterranean water pipeline using satellite-based remote sensing techniques. The primary objective is to elucidate the temporal periodicity of these soil marks and to quantitatively assess the influence of key environmental factors on their visibility. By characterizing the typical patterns and temporal dynamics of soil traces under normal operating conditions, this research intends to establish a robust baseline. This baseline will subsequently form the basis for developing a systematic monitoring methodology. Deviations from this established standard behavior, identified through ongoing remote sensing analysis along the pipeline corridor, will be evaluated as potential indicators of anomalous conditions, thereby contributing to proactive risk assessment and hazard mitigation strategies for pipeline integrity management.

2. MATERIALS AND METHODS

2.1 Description of study area and use case

The Aravissos pipeline, constructed between 1975 and 1978, spans approximately 52 km (51,968 m) from the Aravissos aqueduct to the western embankment of the Gallikos River. Originating at the Aravissos springs, this gravity-fed pipeline, with a diameter of 1650 mm, is made of prestressed concrete reinforced with helical and longitudinal steel. Buried at a depth of 2 to 2.50 meters, it transports water from Mount Paiko to Kalohori and then to the Dendropotamos pumping stations. Serving as a primary conduit for water transportation to Thessaloniki, the pipeline meets approximately 35% of the city's daily water needs. The rupture of the pipeline in 2018 left many areas of the city without regular water supply for three weeks, highlighting its critical importance within the city's extensive water supply network. To evaluate the efficacy of the proposed remote sensing approach, a 1500-meter section of the Aravissos

pipeline with a precisely known location is utilized as a test site. The findings generated by the approach for this segment are compared against the established ground truth to assess its detection accuracy.

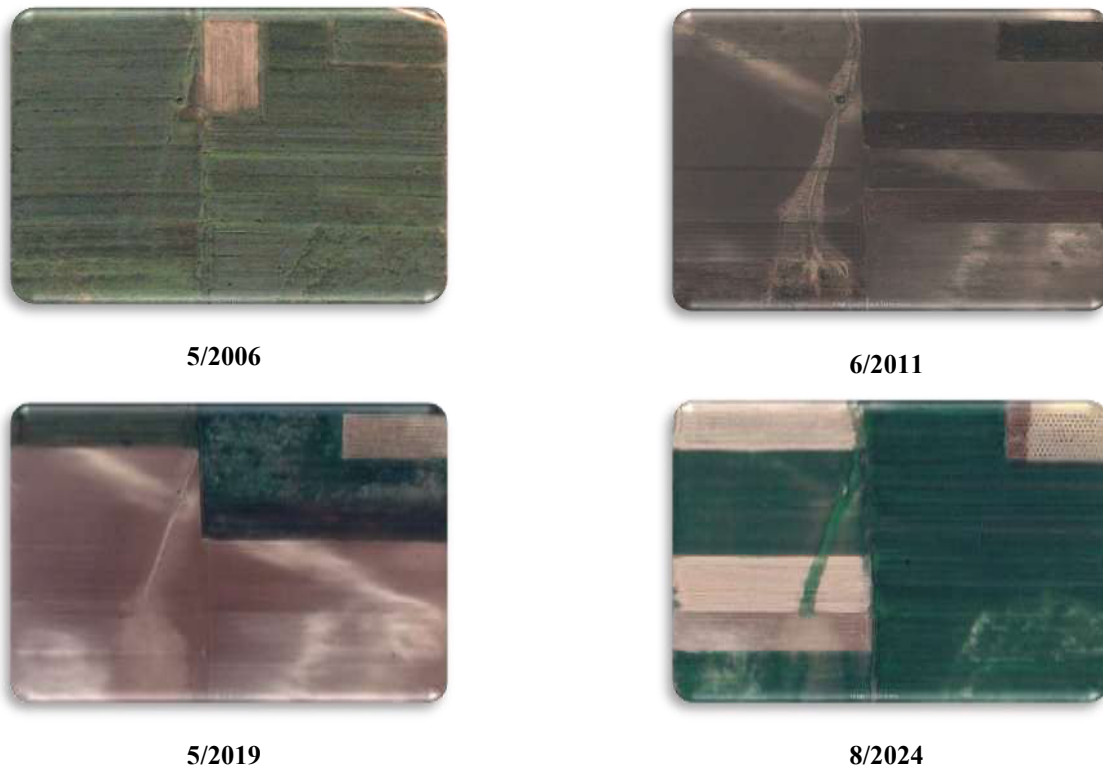


Figure 1. Google Earth imagery examples from different dates (Month/Year indicated below each panel) over the supposed route of the Aravissos pipeline. Surface traces indicative of the pipeline are visible in some instances (e.g., 6/2011) but are less apparent or absent in others (e.g., 5/2006).

2.2 Satellite Monitoring

The satellite data used in this study were obtained from the Sentinel-2 (S-2A) satellite for soil monitoring purposes. The data were accessed from the Copernicus Data Space Ecosystem [7]. Sentinel-2 (S-2A) carries a multi-spectral instrument (MSI) that acquires reflected solar radiance in 13 bands between 440 and 2200 nm spectral wavelength. The spatial resolution is 10 meters for the red, green, blue, and near-infrared bands, 20 meters for six more bands in the NIR (705–865 nm) and short-wave infrared (SWIR, 1600–2200), and 60 meters for three other bands (443, 940, and 1375 nm). Sentinel-2 (S-2A) satellite orbits have a 5-day revisit time in paired operations, with image acquisitions taking place around 9:29 UTC (i.e., 11:29 and 12:29 local time in winter and summer, respectively). The Level-2A processing includes a Scene Classification and an Atmospheric Correction applied to Top of Atmosphere (ToA) Level-1C orthoimage products. The main output of Level-2A processing, which has been used in this study, is an atmospherically corrected orthoimage Surface Reflectance product.

A method was developed to locate the traces on soil that might be related to the underground pipe by analyzing satellite images of the study area and recording the coordinates outlining the soil decolorization. The methodology involved downloading all available images of the region from July 2015 to March 2025, from [7], with a maximum cloud coverage of 10%, totaling 250 images in TIFF format. The method is based on the principle that underground structures can cause the soil above them to change its properties and reflect more light in both visible and infrared spectra. Decolorization marks are not consistently visible, as their formation and shape can vary with weather conditions, soil moisture levels, environmental factors, and the interaction between the soil and the underground pipes. To address this variability, the method involves acquiring multiple satellite images from visible and multispectral spectrum at different times to capture changes in soil moisture and light reflection characteristics. First, multiple georeferenced satellite

images are acquired at different times, capturing changes in soil light reflection characteristics. Following image acquisition, specific analysis transects are established, running both perpendicular to and along the assumed pipeline route. For each date with available satellite imagery, pixel data corresponding to these transects are extracted. These temporal pixel profiles are then arranged chronologically and plotted adjacently, generating composite maps. Such maps provide a systematic visual record of the temporal variations in soil mark appearance, indicative of potential soil surface discolorations, above the pipeline's supposed location.

2.3 Statistical Analysis

The analytical framework processes the multi-temporal satellite imagery by first extracting pertinent spectral information from locations of interest. These locations, defined by user-provided KML files, represent specific sampling points along or near the subterranean structure. For each image in the time series, raw pixel values from selected spectral bands (primarily Red, Green, and Blue) are extracted for these sampling points. To facilitate the visual interpretation of temporal surface changes, two distinct color representation strategies are employed. The first strategy normalizes the extracted pixel colors for each sampling point relative to the overall radiometric range of the entire satellite scene for that specific date and spectral band. This contextual normalization helps in understanding a point's color in relation to its wider surroundings in each image. The second, more targeted strategy aims to enhance subtle chromatic variations observed specifically among the selected set of sampling points for each individual date. This is achieved by first normalizing the pixel values within this subset (i.e., scaling based on the minimum and maximum values found only among the selected points for that date and channel). Subsequently, a non-linear contrast enhancement (i.e. a sigmoid function) is applied to these subset-normalized values. This step amplifies differences, particularly in the mid-range of color tones, thereby improving the visual discernment of subtle soil marks or discolorations that might be indicative of subsurface anomalies. Both resulting color profiles are visualized as time series for each sampling point.

2.4 Drone Imagery Acquisition for Validation

To validate the satellite-derived temporal patterns, high-resolution aerial imagery was acquired using a DJI Mavic 3 drone (DJI, Shenzhen, China) equipped with a 4/3 CMOS sensor (20 MP resolution) and RGB camera. The flight was conducted on 3 April 2024 at approximately 10:51–10:52 local time, under clear conditions with low wind and morning sunlight (sun angle ~30–45°), optimal for enhancing soil tonal contrasts. The drone was flown at an altitude of 100 m above ground level. This drone data serves as a proximal validation proxy, capturing fine-scale soil discolorations also captured in Sentinel-2 imagery and was cross-referenced with contemporaneous satellite data to assess consistency in detected soil marks.

3. RESULTS

3.1 Temporal RGB Color Profiles at Sampling Locations

Figure 2 depicts the spatial configuration of the four KML-defined sampling datasets used in the multitemporal analysis. The white polyline traces the centerline of the Aravissos high-pressure main and itself serves as Dataset 1 (N = 66 regularly spaced points). Three short, colour-coded bars are drawn orthogonally to the pipe to mark the centroid locations of the supplementary datasets: blue for Dataset 2 (N = 15), red for Dataset 3 (N = 106) and green for Dataset 4 (N = 57). Each of those bars corresponds to a small transverse transect from which additional sampling points were generated to capture lateral soil heterogeneity. All base imagery has been deliberately blurred to conceal surrounding land use and thus safeguard critical infrastructure.

Figure 3 presents a comparative visualization of multitemporal Red-Green-Blue (RGB) color profiles for four distinct KML-defined sampling datasets located along the Aravissos pipeline. These profiles correspond to the locations indicated in Figure 2. The figure is organized into four vertically stacked panels, each representing one dataset. The x-axis in all panels denotes the Date, spanning from 2016 to 2024, while the y-axis lists the individual Sampling Points (e.g., LinePoint 1, LinePoint 5, etc.) within each dataset. Each horizontal bar within a panel illustrates the temporal evolution of the RGB color signature for a specific sampling point over the observed period.



Figure 2. Spatial layout of KML-defined sampling areas for multitemporal analysis along the Aravissos pipeline. The primary pipeline route (white line) constitutes the first dataset (N=66 points). Perpendicular markers indicate representative locations for three additional datasets: blue (Dataset 2, N=15), red (Dataset 3, N=106), and green (Dataset 4, N=57). Background imagery is intentionally obscured for security.

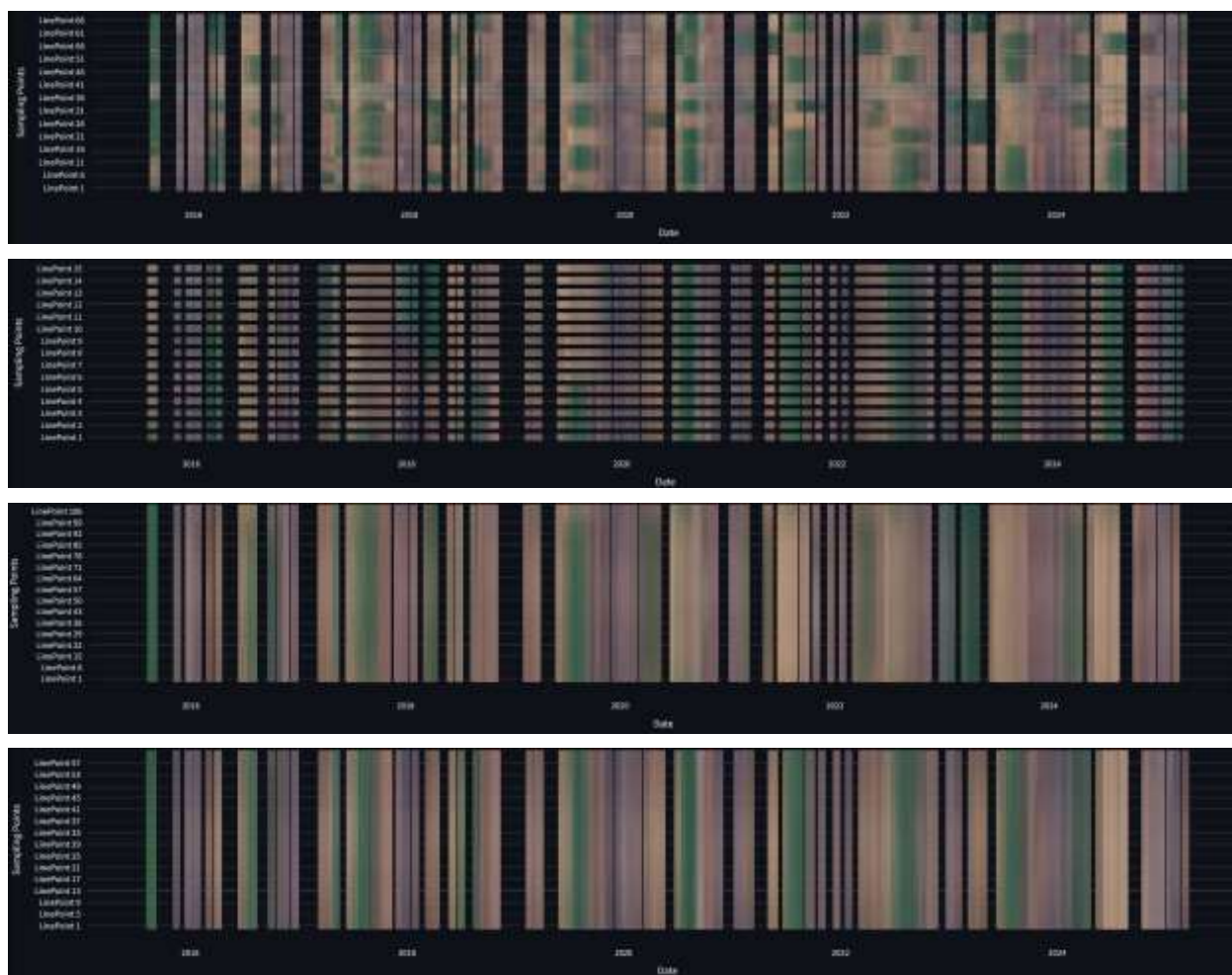


Figure 3. Multitemporal RGB color profiles for the four distinct KML-defined sampling datasets along the Aravissos pipeline, corresponding to the locations indicated in Figure 2. Panels (top to bottom) show temporal RGB profiles for datasets linked to Figure 2 locations: Dataset 1 (white line, N=66), Dataset 2 (blue marker, N=15), Dataset 3 (red marker, N=106), and Dataset 4 (green marker, N=57).

From Figure 3 it becomes obvious that Dataset 1 (top panel), understood to correspond with the route of the underground water pipeline, exhibits clear temporal patterns for each individual sampling point. However, a spatially uniform pattern across the entire set of these sampling points (i.e., along the pipeline's horizontal extent at any single point in time) is not readily apparent from this visualization, suggesting considerable local variability along the route. A notable exception within this dataset is the region encompassing LinePoints 36 through 41; these specific points display a considerably more stable temporal RGB signature, with less pronounced seasonal greening compared to adjacent points, a behavior consistent with the presence of a persistent, non-vegetated feature such as a building.

Furthermore, a prominent and recognizable pattern across all four datasets is the cyclical variation linked to seasonal vegetation changes. This is generally observed as a transition towards greener hues from early spring through mid-summer, indicative of vegetation growth, followed by a shift to browner or more subdued tones during autumn and winter, reflecting vegetation senescence or dormancy. This overarching seasonal dynamic observed in Dataset 1 is also evident in the subsequent three datasets, which are understood to represent different sampling areas or transects related to the pipeline. Beyond such specific, localized features as the aforementioned building-like signature in Dataset 1 or the generally subdued signature of Dataset 2 (which could indicate a specific stable land cover like rock), the figure does not suggest extensive regions exhibiting unexplained, persistently anomalous behavior throughout the observation period. The majority of the temporal variations appear attributable to regular seasonal cycles or episodic events affecting the land surface.

3.2 Analysis of Near-Infrared (NIR) Temporal Profiles at Sampling Locations

Figure 4 presents multitemporal Near-Infrared (NIR) profiles for the four distinct KML-defined sampling datasets along the Aravissos pipeline, mirroring the structure, temporal extent, and sampling locations detailed for the RGB data in Figure 3. This NIR perspective offers complementary information to the visible spectrum data. For shared characteristics such as dataset definitions (Dataset 1, N=66; Dataset 2, N=15; Dataset 3, N=106; Dataset 4, N=57), the interpretation of persistent features (e.g., the building signature at LinePoints 36-41 in Dataset 1), or general seasonal cycles, reference can be made to the preceding description of Figure 3.

The primary distinction observed in these NIR profiles, when compared to the RGB data visualization, is a subtle but noticeable internal color variation within the individual temporal segments that constitute the horizontal bars for each date. Specifically, the central part of these small, colored segments often exhibits a paler or visually distinct hue (predominantly in shades of red and brown) compared to their immediate vertical edges. This characteristic visual pattern is particularly pronounced in Dataset 2. The heightened visibility of this feature in Dataset 2 is related to its specific spatial sampling configuration since this dataset represents a cross-section covering a more extensive spatial area than Datasets 3 and 4. Therefore, the NIR response across such a wider transect could contribute to this more apparent intra-segment color differentiation.

3.3 Visualization of Temporally Enhanced Color Differences at Sampling Locations

Figure 5 displays enhanced multitemporal RGB (left column) and Near-Infrared (NIR) reflectance profiles (right column) for the four established sampling datasets. These visualizations are derived from the application of an image enhancement technique (detailed in Materials and Methods) to the original RGB (Figure 3) and NIR (Figure 4) data. For Dataset 1, which samples along the pipeline's presumed route, the enhancement improves overall data clarity but may not yield a sufficiently continuous or unambiguous signature for precise longitudinal tracing of the pipeline. In contrast, Datasets 2, 3, and 4, representing transects across the pipeline, appear to benefit more substantially from the enhancement. These datasets now exhibit more distinct, albeit localized, spectral-temporal patterns that are interpreted as clearer indicators of the pipeline's presence at their respective intersection points. Consequently, spectral features and differentiations that were previously faint or poorly resolved in the original RGB (Figure 3) and NIR (Figure 4) datasets, respectively, are rendered substantially more conspicuous and interpretable in both the enhanced RGB and NIR profiles of Figure 5 (despite there is no significant difference for the case we are interested in here between RGB and NIR enhanced data).

Further analysis of the enhanced datasets (Figure 5) highlights distinct temporal signatures related to the pipeline. Notably, in Dataset 2, the pipeline's signal becomes clearly discernible only in the period post-2020, after which it appears to exhibit periodic variations in its detectability. Prior to 2020, traces of the pipeline in this particular dataset remain largely indistinct. Conversely, Datasets 3 and 4 demonstrate a pipeline signal characterized by marked temporal

consistency, apparently uninfluenced by seasonal variations throughout the observation period. To rigorously investigate these differentiated behaviors and the nature of these temporal patterns, a statistical analysis is subsequently presented.

3.4 Temporal Statistics of Derived Proxy Metric: Mean and Standard Deviation

Figure 6 presents the temporal profiles of Root Mean Square (RMS) values and standard deviations derived from enhanced Near-Infrared (NIR) spectral data, analyzing the period from March 2015 to March 2025. Specifically, the left panel displays the statistical metrics for Dataset 2 (N=15 points), the right panel displays the corresponding metrics for Dataset 3 (N=106 points). In both panels, the red line represents the RMS value (associated with the left y-axis, which is labeled 'mean value'), and the blue line indicates the standard deviation (associated with the right y-axis). These statistics are calculated from the enhanced NIR profiles of the respective datasets. Additionally, each panel includes the relevant enhanced NIR multitemporal color profile strip for contextual visual reference.

Analysis of the temporal profiles in Figure 6 reveals a distinct annual periodicity in both the Root Mean Square (RMS) and standard deviation values derived from the enhanced Near-Infrared (NIR) data for Datasets 2 and 3. This cyclical behavior, characterized by peaks typically occurring mid-year, strongly correlates with seasonal vegetation phenology. Notably, for Dataset 2, this annual periodicity in both metrics appears more consistently defined post-2020, whereas the

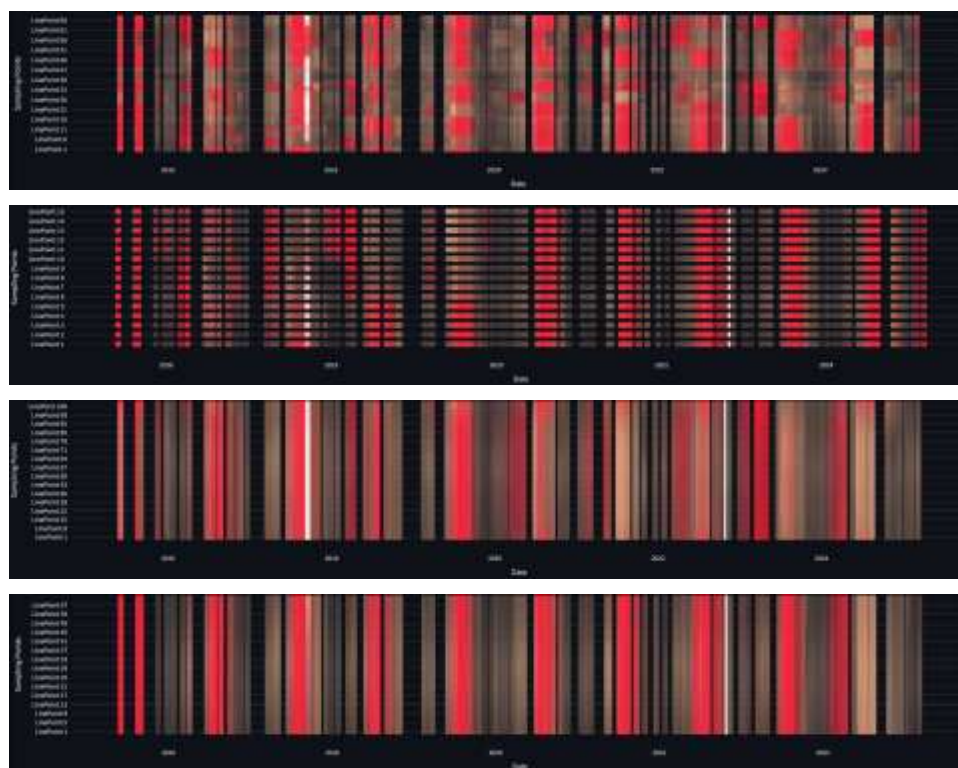


Figure 4. Multitemporal NIR profiles for the four distinct KML-defined sampling datasets along the Aravissos pipeline, corresponding to the locations indicated in Figure 2. Panels (top to bottom) show temporal NIR profiles for datasets linked to Figure 2 locations: Dataset 1 (white line, N=66), Dataset 2 (blue marker, N=15), Dataset 3 (red marker, N=106), and Dataset 4 (green marker, N=57).

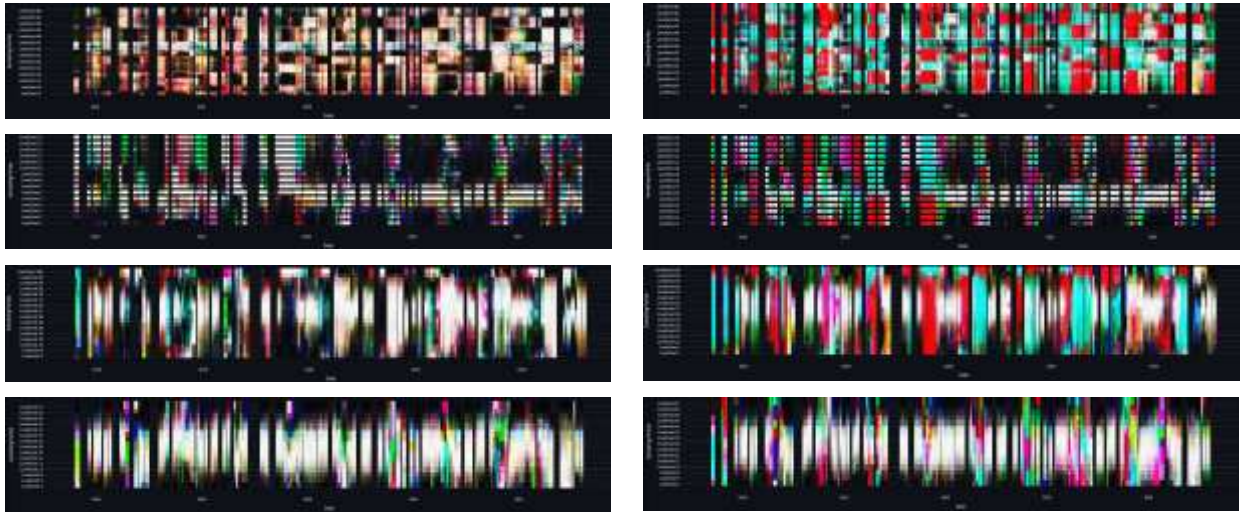


Figure 5. Multitemporal enhanced profiles for the four distinct KML-defined sampling datasets along the Aravissos pipeline, corresponding to the locations indicated in Figure 2. The left column displays the enhanced RGB color profiles, while the right column shows corresponding Near-Infrared (NIR) reflectance profiles. Panels (top to bottom) show temporal RGB profiles for datasets linked to Figure 2 locations: Dataset 1 (white line, N=66), Dataset 2 (blue marker, N=15), Dataset 3 (red marker, N=106), and Dataset 4 (green marker, N=57).

preceding period exhibits greater irregularity, including a significant, anomalous spike in standard deviation in late 2017/early 2018. In contrast, Dataset 3 demonstrates a more robust and temporally consistent annual cycle in both RMS and standard deviation throughout the observation period, with generally higher RMS magnitudes compared to Dataset 2. These observations underscore differing temporal dynamics and spatial heterogeneities between the datasets, warranting further quantitative statistical investigation to ascertain the significance and underlying drivers of these patterns.

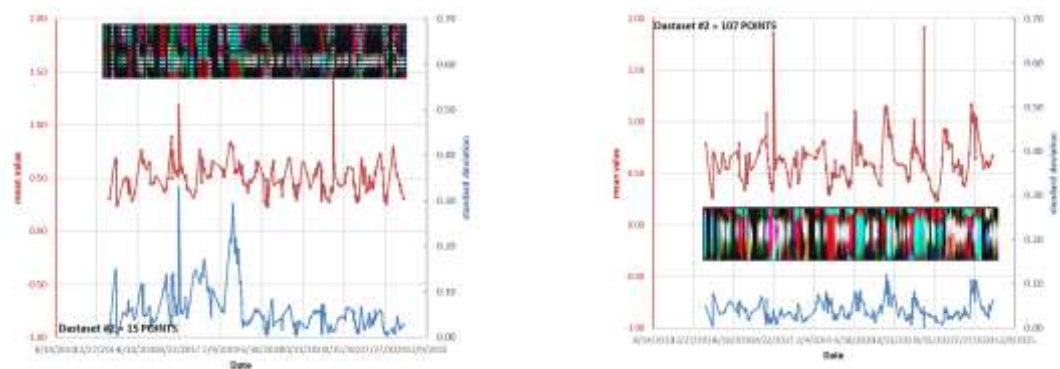


Figure 6. Temporal profiles of mean and standard deviations values for Dataset 2 (N=15 points) and Dataset 3 (N=106 points), derived from the enhanced spectral data corresponding to the period from March 2015 to March 2025 (obtained from NIR dataset).

3.5 Validation via High-Resolution Drone Imagery

Figure 7 presents a high-resolution drone image acquired on 3 April 2024 over a representative section of the Aravissos pipeline route, encompassing areas corresponding to Datasets 2–4 (as defined in Figure 2). The imagery reveals distinct linear soil discolorations manifesting as lighter, elongated strips approximately 20–25 m wide, traversing multiple

agricultural fields. These features exhibit tonal contrasts (lighter brown hues against surrounding plowed soil) consistent with parch marks influenced by subsurface moisture retention differences, as hypothesized in the temporal analysis (Section 1). Notably, the marks align spatially with the enhanced NIR signatures in Datasets 3 and 4, where temporal consistency was observed.



Figure 7. High-resolution drone image (DJI Mavic 3, 3 April 2024) showing soil discolorations along the Aravissos pipeline route. Linear light colored strips indicate potential pipeline traces. The surrounding area has been intentionally blurred for critical infrastructure protection.

4. CONCLUSIONS

The present study successfully developed and applied a multitemporal remote sensing methodology using Sentinel-2 satellite data to identify and characterize soil surface marks associated with the Aravissos subterranean water pipeline. The primary findings demonstrate that targeted image enhancement techniques significantly improved the discernibility of subtle spectral-temporal variations indicative of the pipeline in both RGB and NIR data. While continuous longitudinal tracing of the pipeline via Dataset 1 proved challenging even with enhancement, analysis of transect datasets (Datasets 2, 3, and 4) revealed more distinct, localized indicators of the pipeline's presence. Notably, these surface manifestations, or soil marks, often appear to extend approximately 20-25 meters on either side of the pipeline's central axis. Significant temporal dynamics were also identified: statistical metrics (RMS and standard deviation) derived from enhanced NIR data exhibited annual periodicity strongly correlated with vegetation phenology for Datasets 2 and 3. Furthermore, after enhancement, the pipeline signal in Dataset 2 became clearly discernible with periodic variations only post-2020, whereas in Datasets 3 and 4, the pipeline's signature showed marked temporal consistency, appearing largely uninfluenced by overarching seasonal variations in the surrounding environment. High-resolution drone imagery (DJI Mavic 3, 3 April 2024) provided preliminary validation, confirming linear soil discolorations (20–25 m wide) aligned with enhanced transect datasets and temporal patterns, thereby disentangling pipeline-induced anomalies from seasonal vegetation cycles.

These results underscore the considerable potential of systematic, long-term satellite remote sensing for detecting and monitoring underground infrastructure by analyzing the temporal evolution of subtle surface manifestations. The observed lateral extent of soil marks (20-25 meters) could be attributed to several factors, including the pipeline's long-term influence on local soil hydrology, alterations in soil compaction and thermal properties radiating from the trench and backfill material, or diffuse micro-leakages affecting soil moisture and nutrient distribution. The developed approach, combining specific enhancement strategies with statistical analysis of temporal profiles, offers a promising, non-invasive tool for establishing baseline conditions and subsequently monitoring for anomalies related to pipeline integrity. However, the study also highlights that detectability is influenced by complex pipeline-soil interactions and local environmental conditions. Therefore, these interpretations, particularly regarding the spatial extent and precise causality of the soil marks, necessitate further rigorous investigation. While drone data offer visual corroboration, full

validation requires quantitative validation through extensive in-situ measurements (e.g., soil sampling, moisture analysis, geophysical surveys), refined spatial-spectral analysis, and correlation with detailed pipeline construction and operational records. Future work should also focus on developing robust algorithms for automated detection and expanding the range of spectral and temporal features analyzed to improve the reliability and broader applicability of this methodology in diverse geographical and operational contexts.

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